



Fatigue behaviour of concrete bridge deck slabs reinforced with GFRP bars

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ARTICLE INFO

Article history:

Received 11 September 2009

Accepted 12 June 2010

Available online 26 June 2010

Keywords:

A. Glass fibres

B. Fatigue

B. Strength

D. Mechanical testing

ABSTRACT

GFRP bars are often used for the internal reinforcement of concrete bridge deck slabs as an alternative to traditional steel reinforcements with excellent results in terms of corrosion resistance. Several experiments on bridge decks were conducted to evaluate their structural behaviour but their fatigue performance still needs an adequate experimental investigation. This paper presents the results of an experimental campaign on four full scale concrete bridge deck specimens reinforced with GFRP bars that were designed, constructed and tested to resist cyclic moving loads. Two hydraulic jacks were used to simulate moving concentrated loads. After the cycles, the load was increased to the static failure. The slabs reinforced with GFRP bars showed a better fatigue performance compared to the requests of the European codes.

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1. Introduction

In several cold countries, the temperature fluctuation and the de-icing that is traditionally done with salt along with sand and gravel, often cause the corrosion of steel reinforcing bars and affect the durability of bridge decks. The corrosion of steel rebars causes cracking and spalling of concrete bridge decks with consequent elevated costs for rehabilitation and traffic disruption. Starting from the early nineties the possibility to replace steel rebars with GFRP rebars (that resist fairly well to salt water), as reinforcement of concrete bridge slabs, has been investigated to increase their service life in these unfavourable conditions (see e.g. [1]).

The GFRP bars, compared to the traditional steel rebars, present a lower modulus of elasticity and smaller transverse shear strength. As a consequence, the overall shear capacity of concrete members reinforced with GFRP bars is lower than that of an equivalent traditional member reinforced with the same quantity of steel.

The deck slabs sustain the moving concentrated loads represented by the wheels of the vehicles, and are therefore susceptible to fatigue failures. Often, after cracking, the deck slabs support the traffic loads through arching behaviour. In these cases, compressive membrane forces are generated by the restraining action of the supporting girders. As a consequence, the slab can reach failure by local punching under concentrated loads. Several research projects were carried out to study the effects of the travelling concentrated loads on traditional concrete deck specimens reinforced with steel bars. Okada et al. [2] tested some steel reinforced deck

slabs and found that the formation of wider cracks due to the moving loads caused a reduction of the shear stiffness of the deck. In other experiments conducted on reduced scale models [3] it was shown that the cracking was radial in the case of a single pulsating load while the crack pattern was of a grid-shape in the case of moving loads. In the last case both the flexural and shear fatigue resistance were decreased.

Full scale bridge deck slabs reinforced with GFRP bars were presented in [4]. No bond loss was detected and the crack pattern was typical of bending of the concrete deck spanning between the steel girders. The deck degradation during fatigue was found to be comparable to traditional steel reinforced concrete decks. El-Ragaby et al. [5] tested six full-size GFRP reinforced slabs and found that the punching shear was the only mode of failure under fatigue loads. The GFRP reinforced decks had a better fatigue performance and longer fatigue life compared to those reinforced with steel. Therefore, it can be stated that the behaviour of these structural elements reinforced with GFRP bars and subject to moving cyclic loads is not fully assessed.

The present paper describes experimental tests performed on full scale bridge slab prototypes (5000 × 2480 × 200 mm) reinforced with GFRP bars. The specimens are reinforced with 16 mm diameter GFRP rebars whose external surface is sanded and spirally wounded. The cyclic loads are applied by two hydraulic jacks alternately pulsating.

The mechanical behaviour of the unidirectional glass fibre reinforcing bars was investigated and compared with available experimental data. Bond tests were performed to verify the suitability of the rebars.

The aim of the research is to verify if the rules suggested by the European Codes for these structural elements can be extended to bridge deck slabs reinforced with GFRP rebars and to study their

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bond efficiency across the cracks when the slabs are subject to cyclic, moving loads.

2. Design considerations

The slabs specimens to be tested were designed to be representative of a full scale bridge deck. For the design process reference was made to the European codes ([6–8]) to define both the acting loads and the mechanical properties of concrete. The mechanical properties of the GFRP rebars provided by the producers were adopted for the design process but subsequently tested by the authors.

The sequence of design considerations for these specimens is reported in the following paragraphs.

2.1. Geometry of the specimens

The dimensions of the specimens are described in Fig. 1. In particular, the deck thickness (200 mm) adopted is the most frequent all over the world in stringer bridges and girder and crossbeam systems.

The ratio between the distance of a free edge from the border of the nearest loading device (that simulates the contact area of the wheel of a lorry) and the center-to-center distance of the supports (which simulate the bridge beams) is $2050/1500 = 1.37$. According to elastic thin plate theory (usually referred to as Kirchhoff's plate theory) and the value of this ratio, the free edges of the specimens have very small influence on the behaviour of the deck. Therefore, the outcomes of the experimental tests should be similar to those that could be obtained when adopting a continuous plate. Moreover, the ratio between the center-to-center distance of the supports and the deck thickness, equal to $1500/200 = 7.5$, should minimize the arch effect inside the specimen. The two overhangs, 490 mm long, guarantee a suitable anchorage length of the straight GFRP reinforcing bars and allow an easy road haulage (according to the Italian standards, the specimen becomes an outsize load when larger than 2500 mm).

2.2. Traffic loads

Traffic running on bridges produces a stress spectrum which may cause fatigue. The stress spectrum depends on the geometry of the vehicles, the axle loads, the vehicle spacing, the composition of the traffic and its dynamic effects. The European codes define five fatigue load models. One of them consists of a set of idealized lorries, called frequent lorries. The maximum and minimum stresses should be determined from the most severe effects of different frequent lorries, separately considered, travelling alone along the appropriate lane. Another model consists of a set of standard lorries which together produce effects equivalent to those of typical traffic on European roads. In this model the frequency of each lorry is described through the percentage of each standard lorry in the traffic flow that in turn depend on the route type and on the traffic category of the bridge. All these data are somewhat confusing. Therefore, once stated that the maximum wheel load among all those previously described is 95 kN, it was decided to adopt a 140 kN load (that is a load 1.5 times the maximum one imposed by the codes) to design the specimens. Moreover, the wheel contact area adopted is 200×300 mm, that is a contact area slightly lower than the one given by the codes for a 35 kN wheel and markedly lower than that of a 95 kN twinned wheel. These data depend on the need to reduce the total number of cycles that a specimen can undergo before collapse during the experimental tests.

2.3. Amount and distribution of the GFRP reinforcement

The GFRP rebars adopted are made of unidirectional E-glass fibres. They were manufactured by combining the pultrusion process with a wrapping process. The external surface of the rebar is deformed by wrapping a yarn of fibres along the length of the bar and then bonding quartz sand to the bar surface (see Fig. 2c), in order to guarantee a bond similar to that of a ribbed steel rebar. This process implies a loss of parallelism of the fibres that are placed in the external crown of the circular cross section, and causes a reduction of the tensile strength of the rebar. The mean strength of 16 mm diameter rebars of this kind (three specimens were tested, see Fig. 2a and b and [9]) was 885 MPa with very small scatters while the mean longitudinal elastic modulus was 39 GPa.

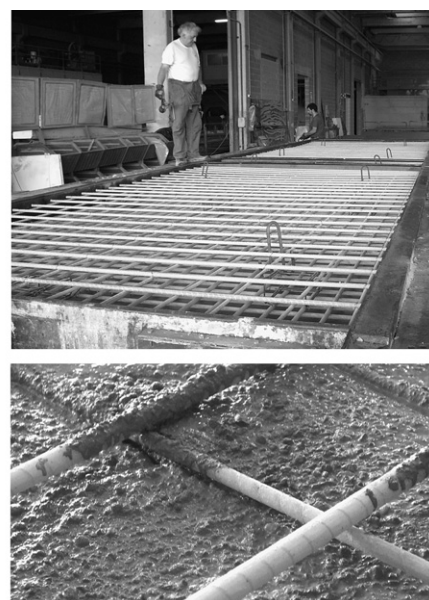
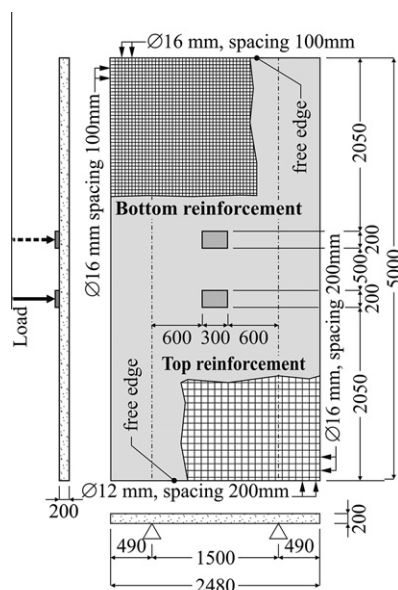


Fig. 1. Specimen dimensions and reinforcement.

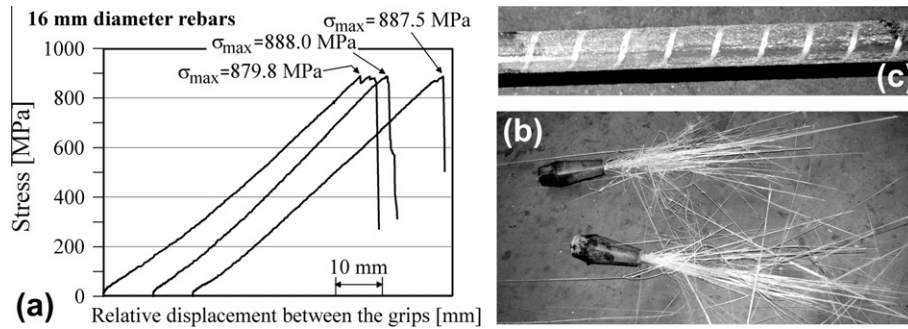


Fig. 2. Tensile tests on 16 mm diameter GFRP rebars: (a) stress vs. grips displacement; (b) tensile failure; (c) rebar external surface.

When comparing these mechanical properties with those of a ribbed steel rebar it has to be stated that tensile strength is 1.5 time higher, whereas the longitudinal elastic modulus is much smaller, 20% with respect to steel. This differences, together with the different behaviour in aggressive environment, imply a change in the design philosophy.

The amount of steel reinforcement is usually determined both to get an adequate load carrying capacity of the slab (under monotonic and cyclic loading) and to limit crack width in concrete under service loads (and hence improve durability of the steel reinforcement), whereas deflection control is not a problem. This is not when dealing with GFRP rebars which resist fairly well to salt water (boats have been made of fibreglass during the last 50 years) and therefore do not need protection. Moreover, their low longitudinal elastic modulus stresses the need to keep low displacements (the limit set is 1/250 of the center-to-center distance of the supports, that is 6 mm) in the cracked concrete slab under the maximum service load (i.e. two 150 kN loads on 400×400 mm contact surfaces placed at mid-span with a center-to-center distance of 1200 mm plus a 9 kN/m^2 distributed load, or a single 200 kN load on a 600×350 mm contact surfaces [6]). In other words, the bottom reinforcement is set to guarantee the durability of the asphalt concrete usually placed over the bridge deck, whereas the same amount of reinforcement is overabundant when referring to the bending strength of the slab.

Once stated that the longitudinal elastic modulus of the GFRP rebar is similar to that of concrete, it is plain that this reinforcement is useless under compression (in other words it cannot help concrete to resist the compression resultant), this is why some guidelines, such as [10], suggest neglecting the contribution in compression of the FRP bars to the flexural capacity. Moreover, GFRP rebars have relatively weak resistance to shear stresses orthogonal to the bar axis and therefore are useless to resist punching too. Nevertheless, in some peripheral zones of the specimen low negative bending moment act during the tests and therefore it was decided to place a top reinforcement equal to 25% of the bottom one in the direction of the supports and 50% of the bottom one in the orthogonal direction (see Fig. 1).

The reason why cyclic loading was not taken into account in the determination of amount and distribution of the GFRP reinforcement comes from the observation that GFRP rebars similar to those adopted to reinforce concrete have oil production application as sucker rods from 1977, in spite of the fact that some guidelines (see [10] and [11]) say that glass-fibres are the FRP bars most susceptible to fatigue.

The fatigue life behaviour of fibre-glass sucker rods differs completely from that of steel rods. The basic difference is that under low service cyclic stress range the fibre-glass rods have a finite life (see [12]). Nevertheless, it has to be mentioned that fibre-glass sucker rods consist of a GFRP body and two steel end fittings (the connecting tools between two composite rods) bonded to both

ends of the rod. It is observed that end fitting forms a critical part in the composite sucker rod systems, and represents the probable cause for the fatigue failure of the system [13]. Referring to sucker rods whose tensile strength is between 620 and 793 MPa (limited by steel end fittings and couplings), the fatigue stress range diagram is the one drawn in Fig. 3 (see [14]). This diagram can be seen as a lower, safe limit for the behaviour of the GFRP rebars, but the maximum stress in the rebars placed into the bridge deck should never exceed 70 MPa (under a 140 kN moving load) because of the limit imposed by bending under service loads, and this should guarantee a vast number of cycles.

Another problem that could affect the behaviour of the specimen is bond of the GFRP rebars under cyclic loading. As far as the authors know nothing has been done yet to investigate this problem. Nevertheless, the acquired knowledge on the behaviour of ribbed steel bars (see [15]) and the low value of the maximum stress in the GFRP rebars allow one to assume that a problem of this kind should not occur in a concrete bridge deck under service loads, whereas nothing can be foretold on the behaviour of the specimen under high cyclic loads.

2.4. Concrete strength

The European codes require compressed concrete verification under cyclic loading by means of the Palmgren–Miner rule:

$$\sum_{i=1}^m \frac{n_i}{N_i} \leq 1 \quad (1)$$

where m is the number of intervals with constant amplitude, n_i is the actual number of constant amplitude cycles in interval i and N_i is the ultimate number of constant amplitude cycles in interval i that can be carried before failure.

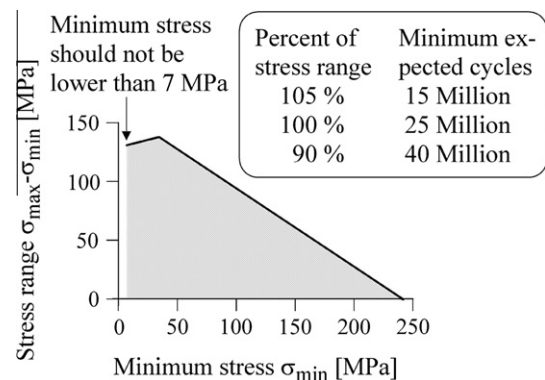


Fig. 3. Maximum stress range of GFRP sucker rods [14].

The ultimate number of constant amplitude cycles (due to the 140 kN load applied on a 200×300 mm contact area moving at mid-span of the specimen) that can be carried before failure is $N_i = 1,138,000$. This number was computed according to the European codes and adopting a concrete C55/67 (see [16]). The mean cylindrical strength of this concrete is 63 MPa and its nominal (design) strength is 36.7 MPa.

Note that it was necessary to choose this type of concrete because when reducing the mean strength of concrete to 53 MPa (concrete C45/55) N_i decreases to 90,000. It was decided not to increase the thickness of the slab to avoid an increase of the dead load on the bridge beams, and because high strength concrete at present is increasingly adopted in common practice.

On the basis of the above data, the design punching shear strength of the specimen (owing to a quasi-static increasing load up to collapse) is 546 kN. This value was computed according to [8] and referring to the minimum shear strength of non-reinforced concrete and takes into account the uncertainties connected to the construction phases (i.e. concrete curing, etc.) by means of a safety factor. Of course, the experimental strength should be higher owing to the lack of significant defects in the specimens.

3. Experimental setup

The experimental setup aims to reproduce as close as possible the working features of a concrete bridge slab. The slabs were constrained by two longitudinal beams (length 5.1 m) to simulate a simple support boundary condition (Fig. 4a). On the top side of the beams two continuous steel half cylinder (diameter 50 mm) were welded to produce the supporting lines (Fig. 4b). Two layers of 5 mm rubber were positioned between the slab and the steel cylinders to prevent local damage of concrete. During the cyclic test of two slabs (named in the following S1, S2) the supports were anchored by means of vertical threaded steel bars to guarantee a

bilateral constrain (see Fig. 4b), as it has to be when the bridge is under service loads. Nevertheless, it was decided to release the vertical steel bars before the static loading, up to collapse, to simulate damaged shear connectors, and hence determine a lower (safe) bound for the residual load carrying capacity of the slabs. Following the same way of reasoning it was decided to remove the vertical steel bars during the cyclic test (see Fig. 4a), with the maximum load, of the third slab (named in the following S3) and during the static test up to collapse of the fourth slab (named S4).

The loading condition of a travelling lorry wheel, according to the European codes, was reproduced by two hydraulic jacks (in the following named A and B) whose axes were perpendicular to the slab longitudinal mid-line. Jacks A and B are equipped with a 1000 kN and 500 kN load cell respectively. The jacks were positioned at a distance of 700 mm (Fig. 1) and equipped with two bottom spherical joints to avoid loads misalignment. Between the slab and the jacks two laminated neoprene prismatic blocks were placed to produce a contact surface of 300×200 mm.

The mid-span displacement was measured using five voltage transducers (LVDTs) placed along the bottom mid-span line of the slab (see Fig. 4c). In particular, LVDT 3 was placed in the mid point of the line, LVDT 2 and LVDT 4 were placed in the center of the contact surface under jack A and jack B respectively. The remaining two LVDTs were at 250 mm from the external edge of the two contact surfaces (LVDT 1 beside jack A and LVDT 5 beside jack B). The recordings of the LVDTs displacements positioned along the mid-span were continuously monitored to investigate the slab deflections during the cyclic loading.

The loading of the slab, to simulate the lorry wheel movement, was defined according to the scheme of a cycle period in Fig. 5. During the first half cycle each jack applied the load following a sinusoidal variation from a minimum value of 5 kN (not representative of the permanent loads on the slab) to the maximum imposed load. In the second half cycle the jacks maintained the minimum value of the load. The phase shift between the two jacks

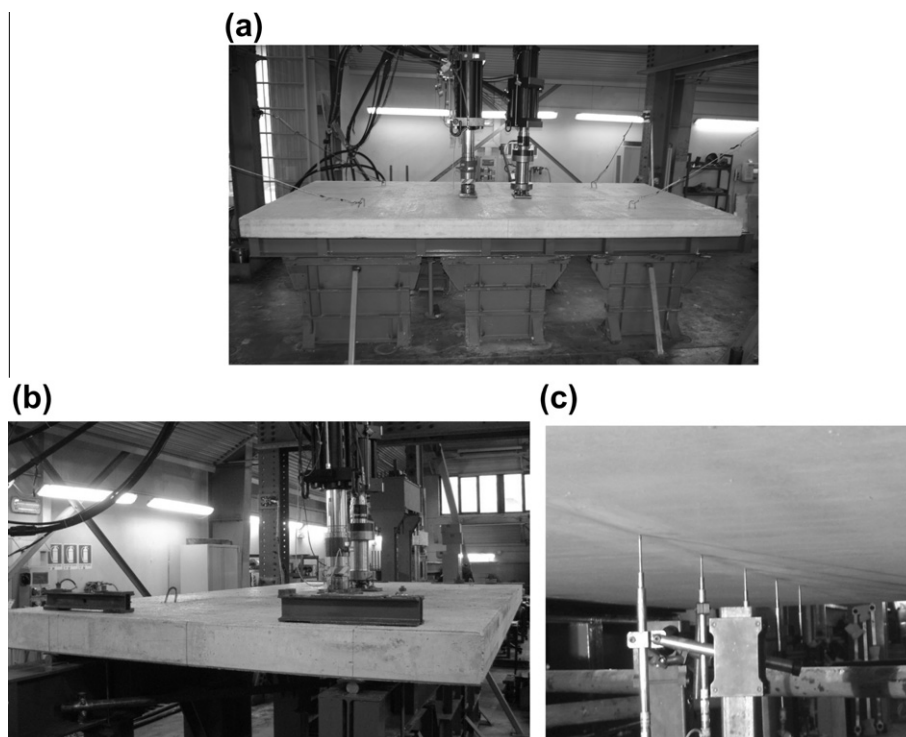


Fig. 4. Experimental setup: (a) slab S3; (b) slab S2; (c) mid-span transducers (LVDTs).

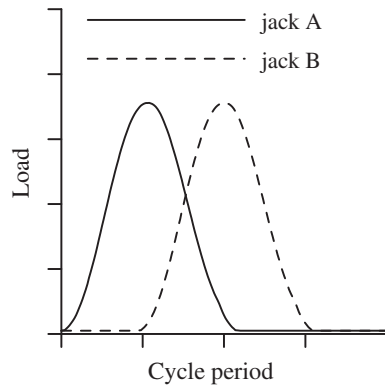


Fig. 5. Shape of the load variation of the two jacks in a cycle period.

Table 1
Summary of the tests features for each sample.

Slab	Test	Max load in cycles (kN)	Frequency (Hz)
S1	Cyclic + static	140	0.75–1.25
S2	Cyclic + static	290	0.7
S3	Cyclic	440	0.2
S4	Static		

was the time of a quarter of cycle in order to have the maximum load of first jack when the second jack is at the minimum. The frequency of the cyclic loads was set according to the maximum load imposed in the range 0.2–1.25 Hz (see Table 1).

4. Experimental results

The experimental campaign involved four slabs. Three slabs (S1, S2 and S3) were cyclically loaded according to the maximum load in the cycle listed in Table 1. Slabs S1 and S2 did not collapse during the cyclic test and therefore were then statically loaded up to failure with a single concentrated load applied by jack A. To observe the residual capacity of the cyclically loaded slabs, the static results were compared to the test of the fourth slab (named S4) not previously cyclically stressed.

Preliminary experimental activities concerned the mechanical characterization of the materials (GFRP rebars and concrete). Tensile tests were performed on 16 mm and 12 mm diameter rebars to have their stiffness and strength used in the slab design (see Fig. 2a for 16 mm diameter rebars). Compressive test of 16 prismatic concrete samples provided an average cylindrical strength of 54.3 MPa

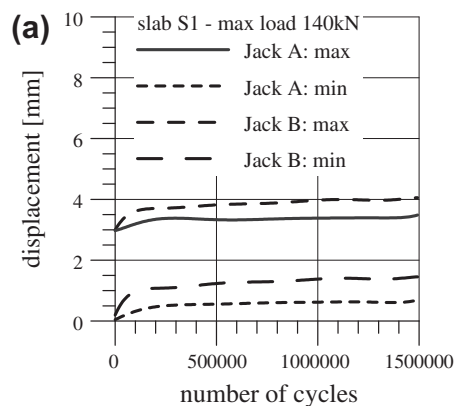
(i.e. 86% of the value adopted to design the slabs) with a standard deviation of 2.5 MPa.

The maximum and minimum displacements of the two jacks recorded in each cycle during the test on slab S1 are reported in Fig. 6a. It should be underlined that the displacement of the jacks are not comparable to the real slab deflection, they contain the influence of the laminated neoprene blocks deformation (see Section 3). The real slab deflection was measured by the LVDTs. During the test the variation of the frequency (0.75, 1 and 1.25 Hz) did not show an influence on the slab response. The test was interrupted after 1.5 million cycles because the slab had reached a stable response as shown by the asymptotic behaviour of the curves in Fig. 6a. The final cracks pattern on the bottom surface of the slab show a radial distribution of the damage from the contact zones (Fig. 6b). The measured displacements by the LVDTs along the mid-line show the influence on the slab response of the damage produced during the cyclic loading. The displacements of the five transducers in the beginning of the test (see Fig. 7a for cycle 225) and after 1.5 million cycles (see Fig. 7b) differ of maximum 0.5 mm. This result suggests that, for the maximum applied load of 140 kN, the slab could work for a number of cycles much greater than the one suggested in the European code. The data in Fig. 7a and b shows that the maximum mid-span deflection of the slab was close to 1.8 mm in the beginning of the test and to 2.5 mm after 1.5 million cycles. This observation confirms the respect of the design criterion on the maximum deflection (1/250 of the center-to-center distance of the supports, that is 6 mm).

The second GFRP reinforced concrete slab (S2) was cyclic loaded with a maximum in the cycle of 290 kN (more than twice the load for sample S1). The diagrams in Fig. 8a show the continuous increase of the jacks displacements during the test. The test was concluded after 140,000 cycles due to the large mid-span displacement reached (≈ 12 mm). After the cyclic test the slab did not fail completely but the damage introduced did not reach a stable state. The crack pattern after the test is reported in Fig. 8b. The influence of the diffuse damage could be observed comparing the mid-span displacements recorded by the LVDTs in the beginning (Fig. 9a) and the end of the test (Fig. 9b). The maximum deflection become more than double with respect the initial one.

The last cyclic test was performed on slab S3, loaded with a maximum force in each cycle of 440 kN (more than three times the one for slab S1). In Fig. 10a the maximum and minimum displacements of the two jacks show a continuous increase up the complete failure of the slab after 403 cycles. The fast damage development (see Fig. 10b) led to the slab collapse for punching of jack B.

The slabs not failed due to the cyclic test (S1 and S2) were re-loaded with an increasing single quasi static load produced by



(b)



Fig. 6. Cyclic test of slab S1: (a) jacks max and min displacements vs. number of cycles; (b) some final cracks below jack B.

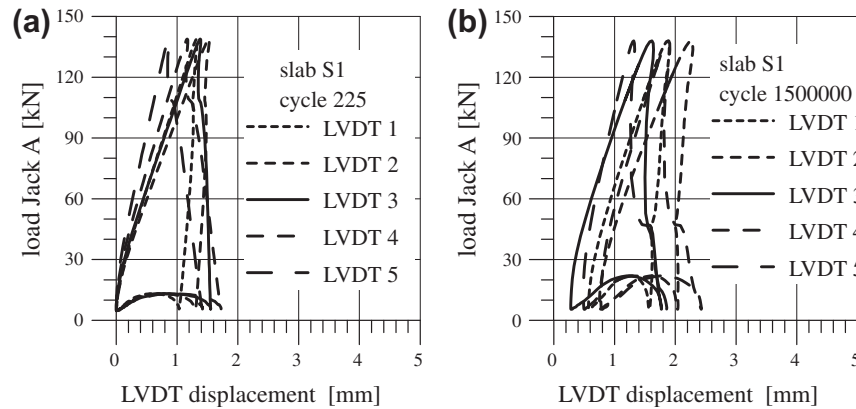


Fig. 7. Cyclic test of slab S1. Load of jack A vs. displacement of the LVDTs: (a) cycle 225; (b) cycle 1,500,000.

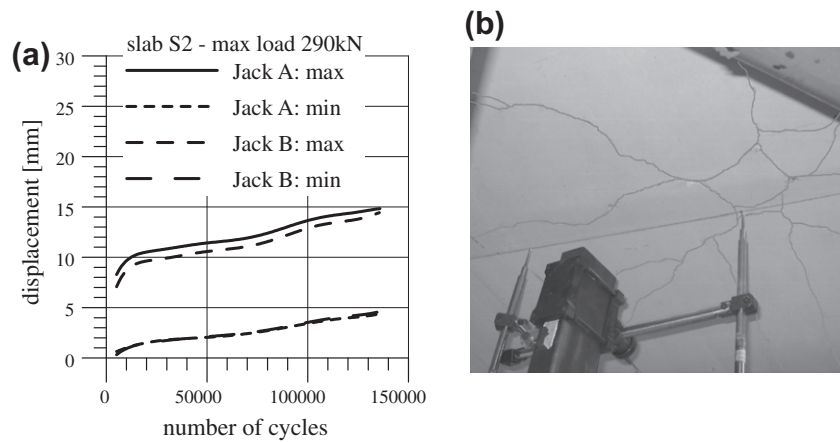


Fig. 8. Cyclic test of slab S2: (a) jacks max and min displacements vs. number of cycles; (b) some final cracks below jack A.

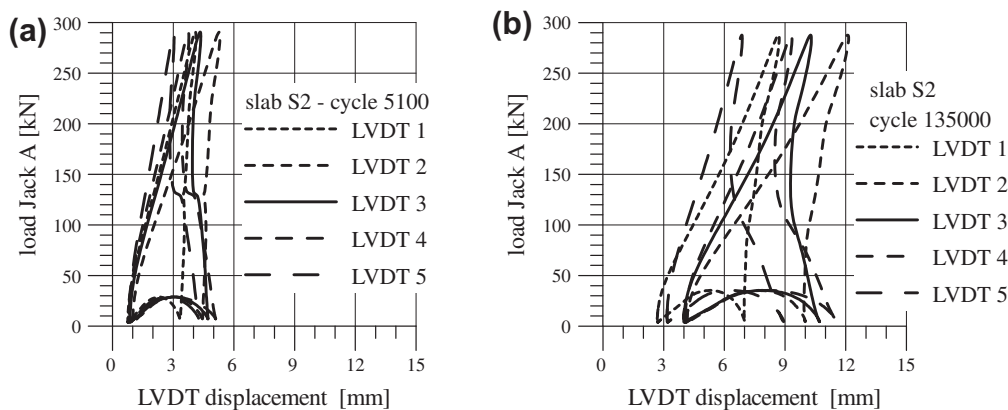
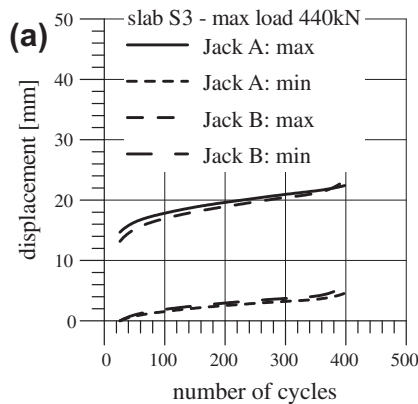


Fig. 9. Cyclic test of slab S2. Load of jack A vs. displacement of the LVDTs: (a) cycle 5100; (b) cycle 135,000.

the jack A at the loading speed of 2 kN per second to compare the residual stiffness of the slabs. The static test consists of two load-unload cycles up to 200 kN and 400 kN and a following ramp up to collapse. In Fig. 11a the static cycles are detailed in terms of load vs. displacement recorded by the LVDT under the jack and compared to the response of slab S4 not previously loaded. The comparison shows that the behaviour of slabs S1 and S4 are very similar with a small reduction of the stiffness of S1, while the slope

of the curve related to slab S2 is markedly lower with similar loading and unloading paths. The difference in the dissipation area of curves in Fig. 11a justifies the large and small damage introduced in sample S2 and S1 respectively during the cyclic test.

Finally the residual capacity in term of load to collapse of the slabs was determined and compared to the one (592 kN) of the undamaged slab S4 (Fig. 11b). The reduction of the maximum static load was of 1.2% and 3.4% for the slab S1 and S2 respectively.



(b)

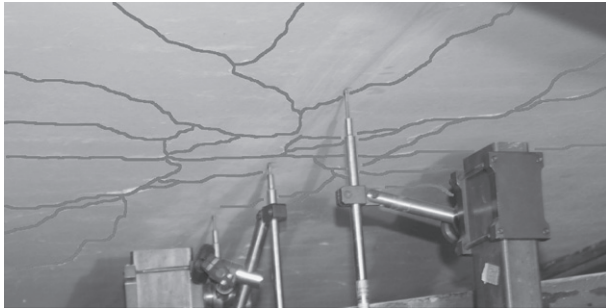


Fig. 10. Cyclic test of slab S3: (a) jacks max and min displacements vs. number of cycles; (b) some cracks below jack A after 150 cycles.

5. Discussion of the experimental outcomes

During cyclic testing of slabs S1 and S2 the displacement of the points placed at mid-span of the free edges of the slabs were measured. The maximum displacement measured in these points was 0.04 mm when dealing with slab S1 and 0.18 mm when testing slab S2, that is something about 0.01 times the jacks max displacement. This outcomes match with the forecasts made in the design phase and therefore give credibility to the similitude with a continuous plate.

When looking at Fig. 6a it can be stated that slab S1 is stiff enough to resist the cyclic load without a significant damage of the asphalt concrete that is usually placed over the bridge deck.

No debonding of the GFRP rebars was observed both under cyclic loading and under quasi static load increase up to ultimate.

After collapse the bars were still bonded to the concrete pieces in the collapse area.

According to the European codes [7] and [8] slab S1 should collapse because of fatigue of concrete after 1.138 million cycles whereas the mechanical behaviour of the slab after 1.5 million cycles was still stable and consistent with a good service behaviour, in spite of the 14% reduction of the experimental concrete strength with respect to the design one. Similarly, slab S2 did not collapse after 140,000 load cycles although the formulas suggested by the European codes foretell collapse of this slab after 1 cycle. If these formulas are replaced with the constitutive law suggested by Karsan and Jirsa [17], that demonstrated to be very reliable at least when dealing with oligocyclic loading (see for instance [18]) we get that collapse of specimen S2 should not occur because of fatigue of concrete, being its maximum stress $\sigma_{c,max} = 29$ MPa, whereas the stability limit (that is the limit under which cyclic loading would not produce any failure of concrete and stress-strain curve would form an hysteresis loop) is 34 MPa. Incidentally, maximum concrete stress in slab S3 is 44 MPa.

The increase of the jacks displacement at the beginning of cyclic loading (i.e. first 100,000 cycles for specimen S1 and first 10,000 cycles for specimen S2, see Figs. 6 and 8) matches with the constitutive law suggested by Karsan and Jirsa [17], which predicts an increase of the strain inside concrete until the stability curve is encountered. The moment when the stability curve is encountered depends on the stress level, i.e. on the value of the applied load and on the position inside the cross section of the slab (see for instance [18]).

The small difference between the load carrying capacity of slab S4 and the residual load carrying capacities of slabs S1 and S2 demonstrates that in these two specimens the strength decrease due to cyclic loading is negligible.

The collapse mechanism in all the static experimental tests was punching. This is usual when dealing with reinforced concrete bridge decks (see for instance [19–23]). In particular, looking at specimen S4 it was observed that under low loading longitudinal cracks formed on the bottom surface of the slab, originating below the load point. As the load increased, cracks formed radially too. At collapse a rectangular crack similar to the shade of the laminated neoprene prismatic block formed on the top surface of the slab (see Fig. 12). These outcomes match the assumptions by Kinnunen and Nylander [19] to describe the deck behaviour.

The experimental punching shear strengths of specimens S1, S2, and S4 (see Fig. 11) are just slightly higher than the one predicted by the European codes [7] and [8] (that takes into account the minimum shear strength of non-reinforced concrete) although significant defects were not found in the specimens.

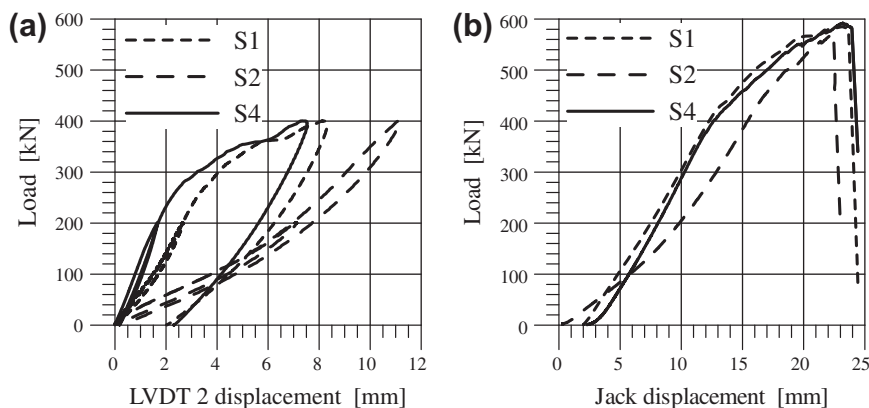


Fig. 11. Static tests: load vs. displacement of (a) LVDT2 and (b) jack.

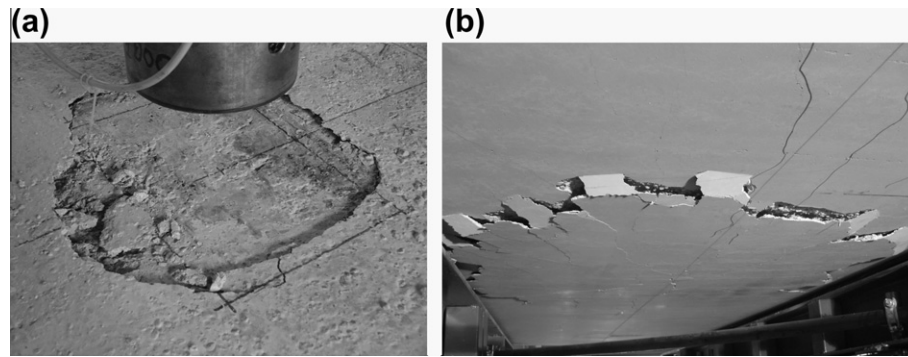


Fig. 12. Static failure mechanism of slab S1: (a) top and (b) bottom view.

6. Conclusions

Four full scale concrete bridge deck specimens reinforced with GFRP bars were designed and tested under cyclic moving loads and under a quasi static load increase up to ultimate. Their experimental behaviour show a better response with respect to the requests of the European codes. Infact, specimen S1 after 1.5 million cycles of a moving load equal to 1.5 times the maximum one imposed by these codes, applied on a contact area that is half of the one suggested in the same codes was still able to resist accomplishing all the specifications to guarantee a correct service limit state, and its load carrying capacity reduced by only 1.2%. Slab S2 after 140,000 cycles, made with a moving load equal to three times the maximum one imposed by these codes (placed on the same reduced wheel contact area) had load carrying capacity reduced by only 3.4%, although the residual stiffness is not compatible with the service limit state.

Under low and high cyclic loads concrete behaved much better than expected, at least if the forecast is made by means of the European codes.

Specimens S1, S2 and S4 collapsed statically because of punching, at a loading level equal to the weight of an heavy lorry placed on a 200×300 mm area, i.e. a load much higher than required.

These outcomes strengthen the belief that concrete bridge decks reinforced with GFRP rebars can represent a good compromise to resist the severe environment.

Acknowledgements

The financial support of the Italian Ministry of the University (Project PRIN 2006), the technical supports of Sireg S.p.A. for the GFRP rebars and of COOPSETTE for the production of the slabs are gratefully acknowledged.

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